SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 30 Minutes

QUESTIONS: 30 **Total Marks:30**

Annexure A

MCQ

Code of Conduct:

I V G ABBIRAMY - 192524097 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V G ABBIRAMY

Annexure A

Circle the most appropriate option for each question.

Q1. Which of the following best describes a Logical Agent in AI?

a) An agent that reacts randomly to the environment c) An agent that only follows pre-defined scripts b) An agent that uses a knowledge base and logical d) An agent that learns from rewards inference to decide actions Answer : b	
Q2. In Propositional Logic, 'P → Q' is logically equivalent to:	
a) P ∧ ¬Q c) ¬Q → ¬P only b) ¬P ∨ Q d) Both b and c Answer : b	
Q3. Conjunctive Normal Form (CNF) requires the formula to be expressed as:	
a) A disjunction of conjunctions c) A conjunction of literals only b) A conjunction of disjunctions (clauses) d) A sequence of implications Answer : b	
Q4. In FOL, the sentence ∀x: Student(x) → Passes(x) means:	
a) Some students pass c) If something passes, it is a student b) Every entity that is a student passes d) There exists at least one student who passes Answer : b	
Q5. Unification in FOL is the process of finding substitutions that:	
a) Convert clauses to CNF c) Prove a goal using backward chaining b) Make two logical expressions syntactically identical d) Derive facts using Modus Ponens Answer : b	
Q6. In the Resolution algorithm, deriving the empty clause indicates:	
a) The knowledge base is consistent c) A contradiction — the original goal is proven b) The negated goal is satisfiable d) More clauses need to be added Answer : c	
Q7. Forward Chaining is best described as a _______ strategy.	
a) Goal-driven / top-down c) Heuristic-based depth-first b) Data-driven / bottom-up d) Probabilistic inference Answer : b	
Q8. Backward Chaining begins reasoning from:	
a) All known facts toward a conclusion c) A random clause in the knowledge base b) The goal and works backward to verify supporting factsd) The resolution refutation tree Answer : b	

Q9. Which step is NOT part of converting a sentence to CNF?

a) Eliminate biconditionals (↔) c) Apply Skolemization to remove existential quantifiers

b) Move negations inward using De Morgan's laws d) Distribute AND over OR

Answer : c

Q10. Knowledge Engineering in FOL involves:

a) Training a neural network on domain-specific data c) Writing heuristic search strategies

c) Defining domain concepts, relations, and axioms formally in FOL d) Encoding reward functions for reinforcement learning

Answer : b

Q11. Modus Ponens states: If P → Q is known and P is true, then:

a) ¬Q is true c) P is uncertain

b) Q must be true d) The KB is inconsistent

Answer : b

Q12. The Most General Unifier (MGU) is the substitution that:

a) Is the most specific possible match c) Eliminates all variables from a clause

b) Makes two terms identical with minimal variable d) Applies resolution to produce CNF binding

Answer : b

Q13. Which inference method is more appropriate when a specific goal is known?

a) Forward Chaining c) Backward Chaining

b) Truth Table method d) A* search

Answer : c

Q14. Propositional Logic differs from First Order Logic because it:

a) Supports probabilistic reasoning c) Is always more expressive

b) Cannot represent objects, relations, or quantifiers — d) Handles uncertainty better only truth values of atomic propositions

Answer : b

Q15. Resolution Refutation proves a goal G by:

a) Directly deriving G from the KB using forward chaining c) Checking G in the truth table

b) Adding ¬G to the KB and deriving a contradiction (empty clause) d) Applying Modus Tollens repeatedly

Answer : b

Q16. Learning from observation in AI refers to:

a) An agent receiving numerical rewards from the environment

c) Acquiring knowledge by following explicit symbolic rules

c) Improving performance by examining examples or experience from the environment

d) Memorizing a fixed dataset without generalization

Answer : b

Q17. Inductive learning is best described as:

a) Deriving specific conclusions from general rules

c) Memorizing training data exactly

c) Generalizing a hypothesis from specific training examples

d) Applying search algorithms to find optimal policies

Answer : b

Q18. In a Decision Tree, what does each internal node represent?

a) A class label / output decision c) A test on an attribute / feature

b) A probability distribution over outcomes d) A training data sample

Answer : c

Q19. Explanation-Based Learning (EBL) differs from inductive learning because it:

a) Requires large amounts of training data to generalize c) Uses prior domain knowledge to explain and generalize a single example

b) Learns only from numeric data using regression d) Applies unsupervised clustering to find hidden patterns

Answer : c

Q20. In statistical learning, a Naive Bayes classifier assumes:

a) All features are conditionally dependent on each other given the class c) All features are conditionally independent given the class label

b) The data must follow a uniform distribution d) Labels are determined by a reward function

Answer : c

Q21. Learning with complete data (no hidden variables) in a Bayesian network involves:

a) Using the EM algorithm to estimate missing values c) Directly computing maximum likelihood estimates from observed data

b) Training a deep neural network with dropout regularization d) Applying reinforcement signals to update weights

Answer : c

Q22. The Expectation-Maximization (EM) algorithm is used when:

a) Training data is fully labeled and no variables are hidden

c) The dataset contains hidden (latent) variables that cannot be directly observed

b) An agent explores an unknown environment using trial and error

d) Backpropagation fails to converge in a neural network

Answer : c

Q23. In a Neural Network, the activation function serves to:

a) Initialize the weights of each neuron to zero

c) Introduce non-linearity to enable learning of complex patterns

b) Normalize the input dataset before training

d) Store previously learned examples in memory

Answer : c

Q24. Which of the following correctly describes backpropagation in a neural network?

a) Weights are updated in the forward pass using input gradients

c) The error is propagated backward from output to input layers to adjust weights

b) Outputs of each layer are stored and reused in later epochs

d) A separate model is trained for each hidden layer independently

Answer : c

Q25. A Support Vector Machine (SVM) — a type of kernel machine — classifies data by:

a) Minimizing the number of training examples used

c) Finding the maximum-margin hyperplane that separates classes

b) Grouping similar data points into clusters using centroids

d) Updating weights using Q-values from an environment

Answer : c

Q26. The kernel trick in kernel machines (SVMs) allows:

a) Reducing the number of support vectors needed

c) Operating in a high-dimensional feature space without explicitly computing coordinates

b) Accelerating gradient descent convergence

d) Replacing hidden layers in a deep neural network

Answer : c

Q27. In Reinforcement Learning, the agent aims to maximize:

a) Classification accuracy on a fixed test set

c) Cumulative reward received over time through interaction with the environment

b) The number of features selected from the training data

d) The size of the neural network hidden layer

Answer : c

Q28. The Q-Learning algorithm in Reinforcement Learning is classified as:

a) A supervised learning algorithm requiring labeled datasets
b) An unsupervised clustering technique using centroids
c) A model-free, off-policy temporal difference learning method
d) A decision-tree-based ensemble classifier

Answer : c

Q29. Overfitting in machine learning occurs when:

a) The model performs poorly on both training and test data
b) The learning rate is set too low during gradient descent
c) The model fits training data too closely and generalizes poorly to new data
d) The training set has more features than training examples

Answer : c

Q30. Which of the following best distinguishes Reinforcement Learning from Supervised Learning?

a) RL uses labeled input-output pairs; SL uses reward signals
b) Both RL and SL require explicit reward functions
c) RL learns via trial-and-error with reward feedback; SL learns from labeled examples
d) RL is always faster to converge than SL on any dataset

Answer : c

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions